LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	26
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Components of hard - iron and soft - iron magnetism
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Components of hard - iron and soft - iron magnetism.
5. Specific Objectives	15.1.10.2 Trainee will be able to explain Components of hard - iron and soft - iron magnetism.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of magnetic component diagrams; Safety: No handling of live electrical components; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft electrical power generation?
(b) Define the term 'magnetism' in the context of aircraft components.
(c) List two types of magnetic materials used in aircraft construction.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Hard Iron Magnetism • Sources of Permanent Magnetism • Characteristics of Hard Iron Components • Impact on Aircraft Compass Deviation • Examples of Hard Iron Sources	L & Ds	PPT, IB	15 min
(b)	• Definition of Soft Iron Magnetism • Sources of Induced Magnetism • Characteristics of Soft Iron Components • Variation with Magnetic Latitude • Examples of Soft Iron Sources	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Hard and Soft Iron • Magnetic Coefficients Overview • Distinction in Deviation Behavior • Role in Compass Correction • Summary of Component Differences	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between hard iron and soft iron magnetism.
(b) How does induced magnetism vary with the aircraft's magnetic latitude?
(c) Identify one source of hard iron magnetism and one source of soft iron magnetism.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of hard and soft iron components will assist in understanding deviation coefficients covered in LP 27.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Deviation coefficients: - Coefficient A, Coefficient B, Coefficient C.	Refer to the master syllabus reference list	LP 27

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min